Exponentials & Polynomials

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • [FRQ] • Target: Laws of Exponents & Fractional Powers

Simplify each expression using exponent and radical laws, assuming $x > 0$ and $y > 0$:

a $\frac{(2x^3y^{-2})^3}{4x^5y^{-4}} \quad \frac{8x^9y^{-6}}{4x^5y^{-4}} = 2x^4y^{-2} = \frac{2x^4}{y^2}$

b $(27x^6)^{2/3} \cdot x^{-1} \quad 9x^4 \cdot x^{-1} = 9x^3$

Apply power-of-a-power $(a^m)^n = a^{mn}$ and quotient $a^m/a^n = a^{m-n}$ laws.

2 [Foundational] • [FRQ] • Target: Conjugates & Simplifying Radicals

Rationalize the denominator and simplify $\frac{12}{3-\sqrt{5}}$ into the form $a + b\sqrt{5}$.

Multiply numerator and denominator by the conjugate $3 + \sqrt{5}$:
$\frac{12(3+\sqrt{5})}{9-5} = \frac{12(3+\sqrt{5})}{4} = 3(3+\sqrt{5}) = 9 + 3\sqrt{5}$.

3 [Foundational] • [FRQ] • Target: Polynomial Multiplication & Degree

Expand and simplify $(2x-3)(x^2+4x-5)$, and state the degree and leading coefficient of the resulting polynomial.

$2x(x^2+4x-5) - 3(x^2+4x-5) = 2x^3+8x^2-10x-3x^2-12x+15 = 2x^3+5x^2-22x+15$. It is a degree-3 (cubic) polynomial with leading coefficient 2.

4 [Foundational] • [FRQ] • Target: Polynomial Remainder Theorem

Use the Remainder Theorem to find the remainder when $P(x) = 2x^3 - 5x^2 + 3x - 7$ is divided by $x - 2$.

By the Remainder Theorem, the remainder equals $P(2) = 2(2)^3 - 5(2)^2 + 3(2) - 7 = 16 - 20 + 6 - 7 = -5$.

5 [Foundational] • [FRQ] • Target: Exponential Growth Modeling

A bacterial culture starts with 400 cells and triples every 2 hours. Write an exponential model $N(t)$ for the population after t hours, and calculate the population after 6 hours.

Since the population triples every 2 hours, $N(t) = 400(3)^{t/2}$. At $t = 6$ hours,
$N(6) = 400(3)^{6/2} = 400(27) = 10{,}800$ cells.

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6 [Foundational] • [FRQ] • Target: Factoring Cubic Polynomials by Grouping

Factor the cubic polynomial $P(x) = x^3 + 3x^2 - 4x - 12$ completely by grouping and list all three real zeros.

Group terms: $x^2(x+3) - 4(x+3) = (x+3)(x^2-4) = (x+3)(x-2)(x+2)$. The real zeros are $x = -3,\ -2, 2$.

7 [Foundational] • [FRQ] • Target: Solving Exponential Equations with Common Bases

Solve the exponential equation $27^{2x-1} = 9^{x+2}$ by rewriting both sides with a common base.

Since $27 = 3^3$ and $9 = 3^2$, rewrite as $(3^3)^{2x-1} = (3^2)^{x+2} \Rightarrow 3^{6x-3} = 3^{2x+4}$. Equate exponents: $6x - 3 = 2x + 4 \Rightarrow 4x = 7 \Rightarrow x = \frac{7}{4} = 1.75$.

Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

8 [Intermediate] • [FRQ] • Target: Half-Life & Exponential Decay

A radioactive isotope has a half-life of 15 days. If a sample initially has a mass of $160\,\text{mg}$, write the decay formula $M(t)$ after t days and find how many days it takes for the sample to decay to $10\,\text{mg}$.

$M(t) = 160\left(\frac{1}{2}\right)^{t/15}$. Setting $M(t) = 10$: $\left(\frac{1}{2}\right)^{t/15} = \frac{10}{160} = \frac{1}{16} = \left(\frac{1}{2}\right)^4 \Rightarrow \frac{t}{15} = 4 \Rightarrow t = 60$ days.

9 [Intermediate] • [FRQ] • Target: Radical Equations & Extraneous Solutions

Solve the radical equation $\sqrt{3x+7} - x = 1$ and verify which candidate root(s) are extraneous.

Isolate the radical: $\sqrt{3x+7} = x + 1$ (requires $x + 1 \geq 0 \Rightarrow x \geq -1$). Square both sides: $3x + 7 = x^2 + 2x + 1 \Rightarrow x^2 - x - 6 = 0 \Rightarrow (x-3)(x+2) = 0$. Testing $x = -2$: $\sqrt{1} - (-2) = 3 \neq 1$ (extraneous). Testing $x = 3$: $\sqrt{16} - 3 = 1$ (valid). Only $x = 3$ is a solution.

10 [Intermediate] • [FRQ] • Target: Factor Theorem & Polynomial Division

Given that $x + 2$ is a factor of $P(x) = 2x^3 + kx^2 - 7x - 6$, find the constant k and factor $P(x)$ completely.

By the Factor Theorem, $P(-2) = 0 \Rightarrow 2(-8) + 4k - 7(-2) - 6 = 0 \Rightarrow -16 + 4k + 14 - 6 = 0 \Rightarrow 4k = 8 \Rightarrow k = 2$. Dividing $2x^3 + 2x^2 - 7x - 6$ by $x + 2$ gives $2x^2 - 2x - 3$.

11 [Intermediate] • [FRQ] • Target: Polynomial Zeros, Multiplicity & Sign Analysis

The cubic polynomial $f(x) = (x+2)(x-1)(x-3)$ is graphed below. State its x-intercepts, its y-intercept, and the intervals on which $f(x) > 0$.

The x-intercepts are $(-2, 0), (1, 0), (3, 0)$, and the y-intercept is $f(0) = (2)(-1)(-3) = 6$, i.e. $(0, 6)$. By sign-chart analysis across the simple roots $-2, 1, 3$, $f(x) > 0$ on $(-2, 1) \cup (3, \infty)$.

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12 [Intermediate] • [FRQ] • Target: Compound Interest & Effective Rate Conversion

An investment of $5{,}000$ dollars earns 8% annual interest compounded quarterly. Write the expression for its value $A(t)$ after t years, and find the effective annual percentage yield (APY) rounded to two decimal places.

With quarterly compounding ($n = 4$), $A(t) = 5000\left(1 + \frac{0.08}{4}\right)^{4t} = 5000(1.02)^{4t}$. Since $(1.02)^4 = 1.08243216$, the effective annual percentage yield is APY $\approx 8.24\%$.

13 [Intermediate] • [FRQ] • Target: Binomial & Cubic Algebraic Identities

If $x - \frac{1}{x} = 4$, compute the exact value of $x^2 + \frac{1}{x^2}$ and $x^3 - \frac{1}{x^3}$.

Square both sides: $\left(x - \frac{1}{x}\right)^2 = x^2 - 2 + \frac{1}{x^2} = 16 \Rightarrow x^2 + \frac{1}{x^2} = 18$. Then $x^3 - \frac{1}{x^3} = \left(x - \frac{1}{x}\right)\left(x^2 + 1 + \frac{1}{x^2}\right) = 4(18 + 1) = 76$.

14 [Intermediate] • [FRQ] • Target: Rewriting Exponential Time Scales

Rewrite the monthly growth model $V(t) = 1200(1.44)^{t/2}$ (where t is in years) in the form $V(t) = 1200(1 + r)^t$ and state the annual percentage growth rate r.

Use exponent laws: $(1.44)^{t/2} = \left((1.44)^{1/2}\right)^t = (1.20)^t$. Therefore $V(t) = 1200(1.20)^t$, which reveals an annual growth rate of $r = 0.20 = 20\%$.

Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • [FRQ] • Target: Linear vs. Exponential Growth Comparison • Cross-Lesson: Topic 1.01 (Linear Models & Rate Comparison)

Compare the linear function $L(x) = 6x + 4$ and the exponential function $E(x) = 2^{x+2}$ at $x = 0, 1, 2, 3, 4$. Find all integer solutions to $L(x) = E(x)$ and state for which x we have $E(x) > L(x)$.

At $x = 0$: $L(0) = 4 = E(0)$. At $x = 1$: $L(1) = 10 > E(1) = 8$. At $x = 2$: $L(2) = 16 = E(2)$. At $x = 3$: $L(3) = 22 < E(3) = 32$. At $x = 4$: $L(4) = 28 < E(4) = 64$. Thus the integer solutions are $x = 0$ and $x = 2$, and $E(x) > L(x)$ for $x < 0$ and $x > 2$.

16 [Challenging] • [FRQ] • Target: Polynomial Division with Quadratic Divisors • Cross-Lesson: Topic 2.01 (Systems of Linear Equations)

A polynomial $P(x)$ leaves a remainder of 5 when divided by $x - 1$ and a remainder of -3 when divided by $x + 3$. Find the linear remainder $R(x) = ax + b$ when $P(x)$ is divided by $(x - 1)(x + 3)$.

Write $P(x) = (x - 1)(x + 3)Q(x) + ax + b$. By the Remainder Theorem, $P(1) = a + b = 5$ and $P(-3) = -3a + b = -3$. Subtracting gives $4a = 8 \Rightarrow a = 2$ and $b = 3$, so the remainder is $R(x) = 2x + 3$.

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17 [Challenging] • [FRQ] • Target: Multiplicity & Tangency of Polynomial Graphs • Cross-Lesson: Topic 3.01 (Quadratic Tangency)

Let $f(x) = (x+1)^2(x-2)$. Find the coordinates of all x- and y-intercepts, identify at which intercept the graph is tangent to the x-axis vs. crosses it, and find $f(3)$.

The x-intercepts are $(-1, 0)$ (multiplicity 2, even, so the graph is tangent to the x-axis and bounces) and $(2, 0)$ (multiplicity 1, odd, so the graph crosses the x-axis). The y-intercept is $(0, f(0)) = (0, -2)$, and $f(3) = (4)^2(1) = 16$.

18 [Challenging] • [FRQ] • Target: Systems of Exponential Equations • Cross-Lesson: Topic 2.01 (Linear Systems)

Solve the system of exponential equations $2^{x+y} = 32$ and $3^{2x-y} = 81$, and evaluate $x^2 + y^2$.

Since $32 = 2^5$ and $81 = 3^4$, equating exponents gives the linear system $x + y = 5$ and $2x - y = 4$. Adding yields $3x = 9 \Rightarrow x = 3$ and $y = 2$. Thus $x^2 + y^2 = 3^2 + 2^2 = 13$.

19 [Challenging] • [FRQ] • Target: Scaling Laws with Rational Exponents • Cross-Lesson: Topic 8.01 (3D Solids) & Topic 5.01 (Percentages)

A cube has surface area $S = 6s^2$ and volume $V = s^3$. Express the volume V as a function of surface area S using rational exponents, and find the percentage increase in V when S is multiplied by 4.

From $s = (S/6)^{1/2}$, we have $V(S) = \left(\frac{S}{6}\right)^{3/2}$. When S is multiplied by 4, the new volume is $4^{3/2}V = 8V$, which is an increase of 700%.

20 [Challenging] • [FRQ] • Target: Dual-Radical Equations • Cross-Lesson: Topic 3.01 (Quadratics)

Find all real solutions to $\sqrt{x+5} + \sqrt{x-3} = 4$.

Move one radical: $\sqrt{x+5} = 4 - \sqrt{x-3}$. Square both sides: $x+5 = 16 - 8\sqrt{x-3} + (x-3) \Rightarrow 8\sqrt{x-3} = 8 \Rightarrow \sqrt{x-3} = 1 \Rightarrow x-3 = 1 \Rightarrow x = 4$. Check: $\sqrt{9} + \sqrt{1} = 3 + 1 = 4$.

Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

21 [Foundational] • [MCQ] • Target: Converting Radicals to Rational Exponents

For $x > 0$, which expression is equivalent to $\sqrt[3]{x^5} \cdot \sqrt{x^3}$?

a. $x^{8/5}$

b. $x^{5/2}$

c. $x^{15/6}$

d. $x^{19/6}$ ✓

Convert to rational exponents and add: $x^{5/3} \cdot x^{3/2} = x^{10/6 + 9/6} = x^{19/6}$.

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22 [Foundational] • [MCQ] • Target: Evaluating Exponential Expressions

Analyze the configuration in which if $3^{2x} = 81$, what is the value of 3^{x+1}?

a. 9

b. 18

c. 27 ✓

d. 84

Since $3^{2x} = (3^x)^2 = 81$ and $3^x > 0$, we have $3^x = 9$. Therefore $3^{x+1} = 3^x \cdot 3^1 = 9 \cdot 3 = 27$.

23 [Foundational] • [MCQ] • Target: Polynomial Subtraction & Coefficient Sums

When $(3x^2 - 4x + 5) - (x^2 + 6x - 2)$ is written in the form $ax^2 + bx + c$, what is the value of $a + b + c$?

a. -1 ✓

b. 3

c. 7

d. 11

Subtract like terms: $2x^2 - 10x + 7$, so $a = 2$, $b = -10$, $c = 7$ and $a + b + c = 2 - 10 + 7 = -1$.

24 [Foundational] • [MCQ] • Target: Exponential Decay Factor

A car purchased for $28{,}000$ dollars depreciates in value by 15% each year. Which function models the car's value $V(t)$, in dollars, t years after purchase?

a. $V(t) = 28000(0.15)^t$

b. $V(t) = 28000(0.85)^t$ ✓

c. $V(t) = 28000(1.15)^t$

d. $V(t) = 28000 - 0.15t$

A 15% annual decrease retains $100\% - 15\% = 85\% = 0.85$ of the value each year, giving $V(t) = 28000(0.85)^t$.

25 [Foundational] • [MCQ] • Target: Solving Basic Radical Equations

Find the real root x that satisfies the radical equation $\sqrt{2x - 3} = 5$.

a. 4

b. 11

c. 14 ✓

d. 28

Square both sides: $2x - 3 = 25 \Rightarrow 2x = 28 \Rightarrow x = 14$.

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26 [Foundational] • [MCQ] • Target: Difference of Cubes Factorization

Which of the following is a factor of $x^3 - 27$?

a. $x + 3$

b. $x^2 - 3x + 9$

c. $x^2 + 3x + 9$ ✓

d. $x^2 - 9$

By the difference-of-cubes identity $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$, $x^3 - 27 = (x - 3)(x^2 + 3x + 9)$.

27 [Foundational] • [MCQ] • Target: Intercepts of Shifted Exponential Functions

What is the y-intercept of the exponential function $f(x) = 5(2)^{x-1} - 3$?

a. $(0, -0.5)$ ✓

b. $(0, 2)$

c. $(0, 5)$

d. $(0, 7)$

Evaluate at $x = 0$: $f(0) = 5(2)^{-1} - 3 = 2.5 - 3 = -0.5$, so the y-intercept is $(0, -0.5)$.

28 [Foundational] • [MCQ] • Target: Identifying Exponential Tables

Which table of values represents an exponential function rather than a linear or quadratic function?

a. $(1, 3), (2, 6), (3, 9), (4, 12)$

b. $(1, 3), (2, 6), (3, 12), (4, 24)$ ✓

c. $(1, 1), (2, 4), (3, 9), (4, 16)$

d. $(1, 5), (2, 7), (3, 11), (4, 17)$

In $(1, 3), (2, 6), (3, 12), (4, 24)$, consecutive y-values have a constant multiplicative ratio of 2, which defines an exponential function $y = 1.5(2)^x$.

Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

29 [Intermediate] • [MCQ] • Target: Exponent Base Unification

Using laws of exponents on the relation $\frac{8^x}{2^y} = 16$, determine the exact value of the exponent difference $3x - y$.

a. 2

b. 3

c. 4 ✓

d. 8

Rewrite $8^x = (2^3)^x = 2^{3x}$ and $16 = 2^4$: $\frac{2^{3x}}{2^y} = 2^{3x-y} = 2^4 \Rightarrow 3x - y = 4$.

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30 [Intermediate] • [MCQ] • Target: Finding Coefficients with the Factor Theorem

When the polynomial $P(x) = x^3 - 4x^2 + kx + 6$ is divided by $x - 3$, the remainder is 0. What is the value of k?

a. -1

b. 1 ✓

c. 2

d. 3

By the Remainder Theorem, $P(3) = 27 - 36 + 3k + 6 = 0 \Rightarrow 3k - 3 = 0 \Rightarrow k = 1$.

31 [Intermediate] • [MCQ] • Target: Interpreting Fractional-Period Exponential Models

The population of a town is modeled by $P(t) = 18000(1.04)^{t/3}$, where t is the number of years after 2020. Which statement best interprets this model?

a. The population increases by 4% every year.

b. **The population increases by 4% every 3 years.** ✓

c. The population increases by 3% every 4 years.

d. The population increases by 1.04% every 3 years.

Because the exponent is $t/3$, the growth factor 1.04 (a 4% increase) is applied once every 3 years.

32 [Intermediate] • [MCQ] • Target: Matching Polynomial Coefficients

If $(ax + 2)(bx - 3) = 12x^2 + cx - 6$ for all x, and $a + b = 7$, what are the two possible values of c?

a. -6 **and** -1 ✓

b. -4 and 1

c. -5 and 5

d. -2 and 2

From $ab = 12$ and $a + b = 7$, $\{a, b\} = \{3, 4\}$. Since $c = 2b - 3a$, if $(a, b) = (3, 4)$ then $c = 8 - 9 = -1$; if $(a, b) = (4, 3)$ then $c = 6 - 12 = -6$.

33 [Intermediate] • [MCQ] • Target: Direct Fractional Power Scaling

Given a positive real base $x > 0$ satisfying $x^{3/4} = 27$, evaluate the fractional power $x^{1/2}$.

a. 3

b. 9 ✓

c. 27

d. 81

Raise both sides to the $\frac{2}{3}$ power: $x^{(3/4)(2/3)} = x^{1/2} = 27^{2/3} = (\sqrt[3]{27})^2 = 3^2 = 9$.

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34 [Intermediate] • [MCQ] • Target: Constructing Polynomials from Roots & Multiplicity

The polynomial $p(x)$ of degree 4 has a double root at $x = -2$, a simple root at $x = 1$, a simple root at $x = 3$, and $p(0) = 12$. What is $p(2)$?

a. -16 ✓

b. -8

c. 8

d. 16

Write $p(x) = a(x+2)^2(x-1)(x-3)$. Since $p(0) = a(4)(-1)(-3) = 12a = 12$, we get $a = 1$. Thus $p(2) = 1(4)^2(1)(-1) = -16$.

35 [Intermediate] • [MCQ] • Target: Asymptotes of Exponential Functions

What is the horizontal asymptote of the graph of $f(x) = 4(0.6)^x - 7$ in the xy-plane?

a. $y = -7$ ✓

b. $y = -3$

c. $y = 0$

d. $y = 4$

As $x \to \infty$, $(0.6)^x \to 0$, so $f(x) \to -7$. The horizontal asymptote is $y = -7$.

36 [Intermediate] • [MCQ] • Target: Combining & Re-Expressing Radicals

Express the sum of surds $\sqrt{50} + \sqrt{18}$ as a single radical $\sqrt{k}$ and find the integer radicand k.

a. 68

b. 96

c. 128 ✓

d. 144

Simplify each radical: $\sqrt{50} + \sqrt{18} = 5\sqrt{2} + 3\sqrt{2} = 8\sqrt{2} = \sqrt{64 \cdot 2} = \sqrt{128}$, so $k = 128$.

37 [Intermediate] • [MCQ] • Target: Sum of Cubes Identity

Using the sum-of-cubes identity with $x^3 + y^3 = 72$ and $x + y = 6$, calculate the product xy.

a. 6

b. 8 ✓

c. 10

d. 12

Use $(x+y)^3 = x^3 + y^3 + 3xy(x+y)$: $6^3 = 72 + 3xy(6) \Rightarrow 216 = 72 + 18xy \Rightarrow 18xy = 144 \Rightarrow xy = 8$.

Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • [MCQ] • Target: Exponential Equations Reducible to Quadratics • Cross-Lesson: Topic 3.01 (Quadratics)

What is the sum of all real solutions to $9^x - 12 \cdot 3^x + 27 = 0$?

a. 3 ✓

b. 4

c. 12

d. 27

Let $u = 3^x$: $u^2 - 12u + 27 = 0 \Rightarrow (u - 3)(u - 9) = 0$. Thus $3^x = 3^1 \Rightarrow x = 1$, or $3^x = 3^2 \Rightarrow x = 2$. The sum of the solutions is $1 + 2 = 3$.

39 [Challenging] • [MCQ] • Target: Log-Linearized Exponential Models • Cross-Lesson: Topic 1.01 (Linear Equations) & Topic 6.01 (Data Modeling)

In a semi-log plot where $\log_{10}(y)$ is plotted on the vertical axis against x on the horizontal axis, the data points lie on the straight line $\log_{10}(y) = 2x + 3$. Which function gives y in terms of x?

a. $y = 1000(100)^x$ ✓

b. $y = 3(10)^{2x}$

c. $y = 100(1000)^x$

d. $y = 2x + 1000$

Exponentiate with base 10: $y = 10^{2x+3} = 10^3 \cdot (10^2)^x = 1000(100)^x$.

40 [Challenging] • [MCQ] • Target: Cubic Vieta's Formulas & 3D Surface Area • Cross-Lesson: Topic 8.01 (3D Solids & Surface Area)

The polynomial $P(x) = x^3 - 6x^2 + 11x - 6$ has three positive integer zeros that represent the lengths of the three edges of a rectangular prism. What is the surface area of the prism?

a. 6

b. 11

c. 22 ✓

d. 36

Let the roots be r_1, r_2, r_3. By Vieta's formulas for a cubic, $r_1r_2 + r_2r_3 + r_3r_1 = 11$. The surface area of a rectangular prism with edges r_1, r_2, r_3 is $2(r_1r_2 + r_2r_3 + r_3r_1) = 2(11) = 22$ (indeed, the roots are $1, 2, 3$).

41 [Challenging] • [MCQ] • Target: Extraneous Root Filtering • Cross-Lesson: Topic 3.01 (Quadratics)

For how many real values of x does $\sqrt{x + 7} = x - 5$ hold?

a. Zero

b. **One** ✓

c. Two

d. Three

Require $x - 5 \geq 0 \Rightarrow x \geq 5$. Square both sides: $x + 7 = x^2 - 10x + 25 \Rightarrow x^2 - 11x + 18 = 0 \Rightarrow (x - 9)(x - 2) = 0$. Only $x = 9 \geq 5$ is valid; $x = 2$ is extraneous. Thus there is 1 real solution.

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42 [Challenging] • [MCQ] • Target: Compound Percentage Scaling Across Multiples of Time • Cross-Lesson: Topic 5.01 (Percentages)

A savings account with continuous-style annual growth grows by 60% over every 4-year period. By what percentage does the account grow over an 8-year period?

a. 120%

b. 140%

c. 156% ✓

d. 160%

Over one 4-year period, the multiplier is 1.60. Over two 4-year periods (8 years), the multiplier is $(1.60)^2 = 2.56$, which corresponds to a $(2.56 - 1) \times 100\% = 156\%$ increase.

43 [Challenging] • [MCQ] • Target: Polynomial Composition & Substitution • Cross-Lesson: Topic 10.01 (Function Composition)

If $P(x)$ is a polynomial such that $P(x^2 + 1) = x^4 + 6x^2 + 8$ for all real x, what is the value of $P(5)$?

a. 24

b. 35

c. 48 ✓

d. 63

Set $x^2 + 1 = 5 \Rightarrow x^2 = 4$. Substitute $x^2 = 4$ into the right-hand side:
$P(5) = (4)^2 + 6(4) + 8 = 16 + 24 + 8 = 48$.

44 [Challenging] • [MCQ] • Target: Coordinate Geometry of Polynomial Secants/Tangents • Cross-Lesson: Topic 1.01 (Linear Equations) & Topic 7.01 (Triangle Area)

What is the area of the triangle formed by the x-axis, the y-axis, and the tangent line to $y = x^2$ at $(0, 0)$ shifted to pass through $(2, 4)$ with slope 4?

a. 1

b. 2 ✓

c. 4

d. 8

The line through $(2, 4)$ with slope 4 has equation $y - 4 = 4(x - 2) \Rightarrow y = 4x - 4$. Its x-intercept is $(1, 0)$ and its y-intercept is $(0, -4)$, forming a right triangle with legs 1 and 4 and area $\frac{1}{2}(1)(4) = 2$.

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45 [Challenging] • [MCQ] • Target: Comparing Large Exponentials via GCD Exponents • Cross-Lesson: Topic 2.01 (Inequalities & Ordering)

If $a = 2^{60}$, $b = 3^{45}$, and $c = 5^{30}$, which inequality correctly orders a, b, and c?

a. $a < c < b$ ✓

b. $a < b < c$

c. $c < a < b$

d. $b < c < a$

Express all three as 15th powers: $a = (2^4)^{15} = 16^{15}$, $b = (3^3)^{15} = 27^{15}$, and $c = (5^2)^{15} = 25^{15}$. Since $16 < 25 < 27$, we have $a < c < b$.

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